Homework_13

May 4, 2026

Probability for Data Science

UC Berkeley, Spring 2026

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1 Homework 13 (Due Monday, May 4th at 5 PM)

[1]:
```
import warnings
warnings.filterwarnings('ignore')

from prob140 import *
from datascience import *
import numpy as np
from scipy import stats

import matplotlib.pyplot as plt
%matplotlib inline
import matplotlib
matplotlib.style.use('fivethirtyeight')
```

1.0.1 Instructions

Questions 1, 2, and 3 can be answered based on Tuesday's lecture (and past content) alone.

We strongly encourage you to attempt Question 1, relying on the provided textbook sections and your own reasoning. If you find yourself stuck, we recommend attending office hours or homework party or posting your questions on Ed. If you're having trouble getting started on problems without the use of AI, we encourage you to follow our recommendation. Your homeworks will generally have two components: a written portion and a portion that also involves code. Written work should be completed on paper, and coding questions should be done in the notebook. Start the work for the written portions of each section on a new page. You are welcome to LaTeX your answers to the written portions, but staff will not be able to assist you with LaTeX related issues.

It is your responsibility to ensure that both components of the homework are submitted completely and properly to Gradescope. **Make sure to assign each page of your pdf to the correct question. Refer to the bottom of the notebook for submission instructions.**

1.0.2 How to Do Your Homework

The point of homework is for you to try your hand at using what you've learned in class. The steps to follow:

- Go to lecture and sections, and also go over the relevant text sections before starting on the homework. This will remind you what was covered in class, and the text will typically contain examples not covered in lecture. The weekly Study Guide will list what you should read.
- Work on some of the practice problems before starting on the homework.
- Attempt the homework problems by yourself with the text, section work, and practice materials all at hand. Sometimes the week's lab will help as well. The two steps above will help this step go faster and be more fruitful.
- At this point, seek help if you need it. Don't ask how to do the problem — ask how to get started, or for a nudge to get you past where you are stuck. Always say what you have already tried. That helps us help you more effectively.
- For a good measure of your understanding, keep track of the fraction of the homework you can do by yourself or with minimal help. It's a better measure than your homework score, and only you can measure it.

1.0.3 Rules for Homework

- Every answer should contain a calculation or reasoning. For example, a calculation such as $(1/3)(0.8) + (2/3)(0.7)$ or `sum([(1/3)*0.8, (2/3)*0.7])`is fine without further explanation or simplification. If we want you to simplify, we'll ask you to. But just $\binom{5}{2}$ by itself is not fine; write "we want any 2 out of the 5 frogs and they can appear in any order" or whatever reasoning you used. Reasoning can be brief and abbreviated, e.g. "product rule" or "not mutually exclusive."
- You may consult others (see "How to Do Your Homework" above) but you must write up your own answers using your own words, notation, and sequence of steps.
- We'll be using Gradescope. You must submit the homework according to the instructions at the end of homework set.

1.1 We will not grade assignments which do not have pages correctly selected for each question.

1.2 1. Predicting Scores

[Your answers to this question should be decimal values or equations with numerical coefficients. For the arithmetic, you are welcome use the code cell at the end of the question. It's just there for your convenience – we won't read it.]

Grades in a class are based on a linear combination of a final exam (worth 50% of the grade), a midterm (worth 30%), and homework (worth 20%). Let the random vector $[F \quad M \quad H]^T$ consist of the final, midterm, and homework scores of a randomly picked student.

Suppose the mean vector of $[F \quad M \quad H]^T$ is $[60 \quad 55 \quad 80]^T$ and the covariance matrix is

$$\begin{bmatrix} 121 & 80 & 10 \\ 80 & 144 & 15 \\ 10 & 15 & 9 \end{bmatrix}$$

Suppose the distribution of $[F \quad M \quad H]^T$ is multivariate normal.

a) Find the distribution of the student's overall score $S = 0.5F + 0.3M + 0.2H$. Read through Ch. 23.1 (Random Vectors) and Ch. 23.2 (Multivariate Normal Vectors) to get started.

Let A be the weights.

$$S = \underbrace{\begin{bmatrix} 0.5 & 0.3 & 0.2 \end{bmatrix}}_{A} \begin{bmatrix} F \\ M \\ H \end{bmatrix}$$

$$\begin{aligned} E[S] &= E[0.5F + 0.3M + 0.2H] \\ &= E[0.5F] + E[0.3M] + E[0.2H] \\ &= 0.5(60) + 0.3(55) + 0.2(80) \\ &= 62.5 \end{aligned}$$

$$\begin{aligned} \Sigma_S &= A\sigma A^T \\ &= 71.37 \end{aligned}$$

Therefore $S \sim Normal(62.5, 71.37)$

b) The instructor wonders whether the final exam score F can just be predicted by a linear function of the random variable $X = 0.3M + 0.2H$. Explain whether the least squares predictor of F based on X is linear and why. Then find the least squares predictor of F based on X. Relevant background can be found in Ch. 24.1.2 (Slope and Intercept of the Regression Line) and Ch. 24.3 (Regression and the Bivariate Normal).

Yes, the final exam score can be predicted by a linear function. X is a linear combination of a multivariate normal distribution. Therefore, $\begin{bmatrix} F & X \end{bmatrix}^T$ is a bivariate normal distribution as well.

We showed in lecture that the least squares predictor of a bivariate distribution is linear. Therefore, since $\begin{bmatrix} F & X \end{bmatrix}^T$ are bivariate normal, then the least squares predictor is linear.

$X \sim Normal(32.5, 15.12)$

(This used the code cell to calculate the values)

The least squares predictor of F based on X is thus:

$$\begin{aligned} E[F \mid X] &= E[\sigma_F F_{su} + \mu_F \mid X_{su}] \\ &= \sigma_F E[F_{su} \mid X_{su}] + \mu_F \\ &= \sigma_F \rho \left(\frac{X - \mu_X}{\sigma_X} \right) + \mu_F \\ &= \rho \frac{\sigma_F}{\sigma_X} (X - \mu_X) + \mu_F \end{aligned}$$

$$\begin{aligned} \rho \frac{\sigma_F}{\sigma_X} &= \frac{\mathrm{Cov}(X, F)}{\mathrm{Var}(X)} \\ &= \frac{\mathrm{Cov}(0.3M, F) + \mathrm{Cov}(0.2H, F)}{\mathrm{Var}(0.3M + 0.2H)} \\ &= \frac{0.3\,\mathrm{Cov}(M, F) + .2\,\mathrm{Cov}(H, F)}{0.3^2\,\mathrm{Var}(M) + 0.2^2\,\mathrm{Var}(H) + 2\,\mathrm{Cov}(0.3M, 0.2H)} \\ &= \frac{.3(80) + .2(10)}{0.3^2 \cdot 144 + 0.2^2 \cdot 2 \cdot 0.3 \cdot .2\,\mathrm{Cov}(M, H)} \\ &= \frac{.3(80) + .2(10)}{0.3^2 \cdot 144 + 0.2^2 \cdot 9 + .12 \cdot 15} \\ &= \frac{26}{15.12} \\ &= 1.72 \end{aligned}$$

$$E[F \mid X] = 1.72X + 4.1$$

c) Find the root mean squared error of the predictor in Part **b**. Utilize the equation in Ch. 24.3.1 (Prediction Error) to find the MSE, but remeber that this equation is for when X and Y are standard bivariate normal. This is not the case for this question, so the setup will be slightly different. Additionally, RMSE is the square root of the MSE.

$$\begin{aligned} MSE(E[F \mid X]) &= E[(F - E[F \mid X])^2] \\ &= E[E[(F - E[F \mid X])^2 \mid X]] \\ &= E[\mathrm{Var}(F \mid X)] \end{aligned}$$

$$
\begin{aligned}
\mathrm{Var}(F \mid X) &= \mathrm{Var}(\sigma_X \left(\rho \left(\frac{f - \mu_F}{\sigma_F}\right) + \sqrt{1-\rho^2} Z\right) + \mu_X) \\
&= \mathrm{Var}(\sigma_X \left(\sqrt{1-\rho^2} Z\right)) \\
&= \sigma_X^2 (1-\rho^2)
\end{aligned}
$$

$$
E[\mathrm{Var}(F \mid X)] = \sigma_X^2 (1-\rho^2)
$$

RMSE

[ ]:
```
weight = np.matrix([0.5, 0.3, 0.2])
mu = np.matrix([60, 55, 80])
cov = np.matrix([[121, 80, 10],
      [80, 144, 15],
      [10, 15, 9]])

...
```

1.3 2. Normal Sample Mean and Sample Variance, Part 1

Let $X_1, X_2, \ldots, X_n$ be i.i.d. with mean μ and variance σ^2. Let

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$$

denote the sample mean. In Homework 7, you constructed a random variable

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2$$

called the sample variance. Before proceeding, please review Homework 7, Question 3 in its entirety.

a) For $1 \leq i \leq n$ let $D_i = X_i - \bar{X}$. Find $Cov(D_i, \bar{X})$.

$$\begin{aligned}
\text{Cov}(D_i, \bar{X}) &= \text{Cov}(X_i, \bar{X}) + \text{Var}(\bar{X}) \\
&= \frac{\sigma^2}{n} - \frac{\sigma^2}{n}, \text{by HW 7} \\
&= 0
\end{aligned}$$

b) Now assume in addition that $X_1, X_2, \ldots, X_n$ are i.i.d. normal (μ, σ^2). What is the joint distribution of $\bar{X}, D_1, D_2, \ldots, D_{n-1}$? Explain why D_n isn't on the list.

D_n isn't on the list because it is fully explained by the other variables. In other words, it is

c) True or false (justify your answer): The sample mean and sample variance of an i.i.d. normal sample are independent of each other.

[ ]:

```
```

1.4 3. Normal Sample Mean and Sample Variance, Part 2

a) Let R have the chi-squared distribution with n degrees of freedom. What is the mgf of R?

$R \sim gamma(\frac{n}{2}, \frac{1}{2})$

$$M(t) = \left(\frac{\frac{1}{2}}{\frac{1}{2} - t} \right)^{\frac{n}{2}}$$

b) For R as in Part (a), suppose $R = V + W$ where V and W are independent and V has the chi-squared distribution with $m < n$ degrees of freedom. Can you identify the distribution of W? Justify your answer.

We can express the MGF of R in terms of V and W. We can equate the exponents to make the equality true.

$$M_R(t) = M_V(t) M_W(t)$$
$$\left(\frac{\frac{1}{2}}{\frac{1}{2} - t} \right)^{\frac{n}{2}} = \left(\frac{\frac{1}{2}}{\frac{1}{2} - t} \right)^{\frac{m}{2}} \cdot \left(\frac{\frac{1}{2}}{\frac{1}{2} - t} \right)^{\frac{n-m}{2}}$$

Therefore, $W \sim gamma(\frac{n-m}{2}, \frac{1}{2})$

c) Let $X_1, X_2, \ldots, X_n$ be any sequence of random variables and let $\bar{X} = \frac{1}{n} \sum_{i=1}^{n} X_i$. Let α be any constant. Prove the *sum of squares decomposition*

$$\sum_{i=1}^{n} (X_i - \alpha)^2 \;=\; \sum_{i=1}^{n} (X_i - \bar{X})^2 \;+\; n(\bar{X} - \alpha)^2$$

$$
\begin{aligned}
\sum_{i=1}^{n}(X_i-\alpha)^2 &= \sum_{i=1}^{n}(X_i-\bar{X}+\bar{X}-\alpha)^2 \\
&= \sum_{i=1}^{n}\left[(X_i-\bar{X})+(\bar{X}-\alpha)\right]^2 \\
&= \sum_{i=1}^{n}(X_i-\bar{X})^2+\sum_{i=1}^{n}(\bar{X}-\alpha)^2+\sum_{i=1}^{n}2(X_i-\bar{X})(\bar{X}-\alpha) \\
&= \sum_{i=1}^{n}(X_i-\bar{X})^2+n(\bar{X}-\alpha)^2+2(\bar{X}-\alpha)\sum_{i=1}^{n}(X_i-\bar{X}) \\
&= \sum_{i=1}^{n}(X_i-\bar{X})^2+n(\bar{X}-\alpha)^2+2(\bar{X}-\alpha)\left[\sum_{i=1}^{n}X_i-n\bar{X}\right] \\
&= \sum_{i=1}^{n}(X_i-\bar{X})^2+n(\bar{X}-\alpha)^2+2(\bar{X}-\alpha)\left[\sum_{i=1}^{n}X_i-n\frac{1}{n}\sum_{i=1}^{n}X_i\right] \\
&= \sum_{i=1}^{n}(X_i-\bar{X})^2+n(\bar{X}-\alpha)^2
\end{aligned}
$$

d) Now let $X_1, X_2, \ldots, X_n$ be i.i.d. normal with mean μ and variance $\sigma^2 > 0$. Let S^2 be the "sample variance" defined by

$$
S^2 \;=\; \frac{1}{n-1}\sum_{i=1}^{n}(X_i-\bar{X})^2
$$

Find a constant c such that cS^2 has a chi-squared distribution. Provide the degrees of freedom.

[Use Parts (b) and (c) as well as the result of the previous exercise.]

$$
\begin{aligned}
\sum_{i=1}^{n}(X_i-\mu)^2 &= \sum_{i=1}^{n}(X_i-\bar{X})^2 + n(\bar{X}-\mu)^2 \\
\sum_{i=1}^{n}(X_i-\mu)^2 &= (n-1)S^2 + n(\bar{X}-\mu)^2 \\
\sum_{i=1}^{n}\left(\frac{X_i-\mu}{\sigma}\right)^2 &= \frac{n-1}{\sigma^2}S^2 + \left(\frac{n(\bar{X}-\mu)}{\sigma}\right)^2 \\
\sum_{i=1}^{n} Z^2 &= \frac{n-1}{\sigma^2}S^2 + \left(\frac{\bar{X}-\mu}{\sigma/n}\right)^2 \\
\sum_{i=1}^{n} Z^2 &= \frac{n-1}{\sigma^2}S^2 + Z^2 \\
\underbrace{\sum_{i=1}^{n}(Z^2)}_{gamma(\frac{n}{2},\frac{1}{2})} - \underbrace{Z^2}_{gamma(\frac{1}{2},\frac{1}{2})} &= \frac{n-1}{\sigma^2}S^2
\end{aligned}
$$

$c = \frac{n-1}{\sigma^2}S^2$ and The LHS is gamma$(\frac{n-1}{2}, \frac{1}{2})$, therefore the degrees of freedom is $n-1$

1.5 4. Multiple Regression: Residuals

This exercise assumes the multiple regression model of Section 25.4 of the textbook and uses the same notation as in that section.

a) Find the distribution of the parameter estimate vector $\hat{\boldsymbol{\beta}}$. Derive the answer; don't just quote a result from lecture.

From lecture we established that:

$$Y = X\beta + \epsilon$$
$$Y = X\hat{\beta} + e$$

Which means that the true Y is our estimated signals and some estimated noise. We also established that Y is multivariate normal with mean vector $X\beta$, and covariance matrix $\sigma^2 I_n$, because we assume errors are multivariate normal.

We also established that if $\mathbb{X}$ is full rank, then $\hat{\beta} = (X^T X)^{-1} X^T Y$. $(X^T X)^{-1} X^T$ is a linear transformation matrix, and we can call this A.

Then since we know that $Y \sim \text{Multivariatenormal}(X\beta, \sigma^2 I_n)$, then under the linear transformation rules for Multivariatenormals, $\hat{\beta} \sim \text{multivariatenormal}(AX\beta, \sigma^2 A I_n A^T\ )$

Expanding it out:

$$\begin{aligned} AX\beta &= (X^T X)^{-1} X^T X\beta \\ &= \beta \end{aligned}$$

$$\begin{aligned} \sigma^2 A I_n A^T &= \sigma A A^T \\ &= \sigma^2 (X^T X)^{-1} X^T X (X^T X)^{-1^T} \\ &= \sigma^2 (X^T X)^{-1} \end{aligned}$$

The distribution of $\hat{\beta}$ is $\text{MultivariateNormal}(\beta, \sigma^2 (X^T X)^{-1})$

b) The regression estimate $\hat{\mathbf{Y}}$ can be written as $\mathbf{HY}$ for an $n \times n$ matrix $\mathbf{H}$. This matrix is called the *hat matrix*, probably because it ''puts the hat on $\mathbf{Y}$." Write $\mathbf{H}$ in terms of $\mathbf{X}$. Is $\mathbf{H}$ symmetric?

$$\hat{Y} = HY$$

$$\begin{aligned} \hat{Y} &= X\hat{\beta} \\ &= X(X^T X)^{-1} X^T Y \end{aligned}$$

$$H = X(X^TX)^{-1}X^T$$

$$\begin{aligned} H^T &= \left(X(X^TX)^{-1}X^T\right)^T \\ &= X^{T^T}(X^TX)^{-1^T}X^T \\ &= X^T(X^TX)^{-1}X^T \end{aligned}$$

Therefore $H = H^T$, making it symmetric **c)** Show that **H** is idempotent for **H** defined in Part (b). (If you haven't seen that term before, look it up.)

Idempotent is if H is applied multiple times, it is the same operation. In simpler terms, $H^n = H$

$$\begin{aligned} H^2 &= X\underbrace{(X^TX)^{-1}X^TX}_{\mathbb{I}}(X^TX)^{-1}X^T \\ &= X(X^TX)^{-1}X^T \end{aligned}$$

d) Find the distribution of the residual vector **e**.

$$e = Y - \hat{Y}$$

$$\begin{aligned} e &= Y - HY \\ &= (I - H)Y \end{aligned}$$

e is multivariatenormal$((I - H)\mu_Y, (I - H)\Sigma_Y(I - H)^T\ )$

e) Show that the covariance matrix of **e** is $\sigma^2(\mathbf{I} - \mathbf{H})$.

$$\begin{aligned} (I - H)\Sigma_Y(I - H)^T &= (I - H)\sigma^2 I_n(I - H)^T \\ &= \sigma^2(I - H)(I - H)^T \\ &= \sigma^2(I - H)(I^T - H^T) \\ &= \sigma^2(I - H)(I - H), \text{by symmetric matricies} \\ &= \sigma^2(I^2 - IH - HI + H^2) \\ &= \sigma^2(I^2 - H - H + H^2) \\ &= \sigma^2(I - 2H + H), \text{by idempotent H} \\ &= \sigma^2(I - H) \end{aligned}$$

1.6 Submission Instructions

Many assignments throughout the course will have a written portion and a code portion. Please follow the directions below to properly submit both portions.

1.6.1 Written Portion

- Scan all the pages into a PDF. You can use any scanner or a phone using applications such as CamScanner. Please **DO NOT** simply take pictures using your phone.
- Please start a new page for each question. If you have already written multiple questions on the same page, you can crop the image in CamScanner or fold your page over (the old-fashioned way). This helps expedite grading.
- It is your responsibility to check that all the work on all the scanned pages is legible.
- If you used LaTeX to do the written portions, you do not need to do any scanning; you can just download the whole notebook as a PDF via LaTeX.

1.6.2 Code Portion

- Save your notebook using `File > Save and Checkpoint`.
- Generate a PDF file using `File > Download As > PDF via LaTeX`. This might take a few seconds and will automatically download a PDF version of this notebook.
  - If you have issues, please post a follow-up on the general Homework 13 Ed thread.

1.6.3 Submitting

- Combine the PDFs from the written and code portions into one PDF. Here is a useful tool for doing so.
- Submit the assignment to Homework 13 on Gradescope.
- **Make sure to assign each page of your pdf to the correct question.**
- **It is your responsibility to verify that all of your work shows up in your final PDF submission.**

If you are having difficulties scanning, uploading, or submitting your work, please read the Ed Thread on this topic and post a follow-up on the general Homework 13 Ed thread.